Contents

ENGY 680 Tutorial: Setting Up Your Python + GitHub Environment

This tutorial covers the steps needed to get your computer ready for Urban Energy and Climate problem sets: installing Python (via conda / Anaconda), installing GitHub Desktop, cloning your personal course repo, running a Jupyter notebook, and committing + pushing your work back to GitHub.

Who this is for: students with no prior command-line, Python, or Git experience. Every step works on both Mac and PC (Windows) — where the steps diverge, look for the 🍎 Mac / 🪟 PC labels.

We will practice these steps in class together. Before the first class, just pay attention to Step 0, though if you have time, you may want to get a headstart on installations to save WiFi bandwidth in class.

Note that these steps are primarily tested on a Mac. Behavior on Windows is somewhat dependent on software versions and thus unpredictable. If you encounter persistent problems, reach out to the instructor, TA, or other students to see if there may be a platform-specific challenge or error in these instructions.

Attribution

Claude Sonnet 5 (with reasoning) was used to draft this tutorial from a description of the context, links to the README and homework instructions, and an outline of the 7 major steps: see conversation via Kagi.

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0. Before you start

- You have a github.com account.
- You've emailed the instructor your GitHub username (or accepted the emailed repo invite), so your private repo `[YOUR_USERNAME]/engy680` exists and you have access.
- You have ~2 GB of free disk space (if installing Miniconda) or ~10 GB (if installing the full Anaconda with graphical Navigator) and a decent internet connection.

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1. Install conda (Miniconda) or Anaconda

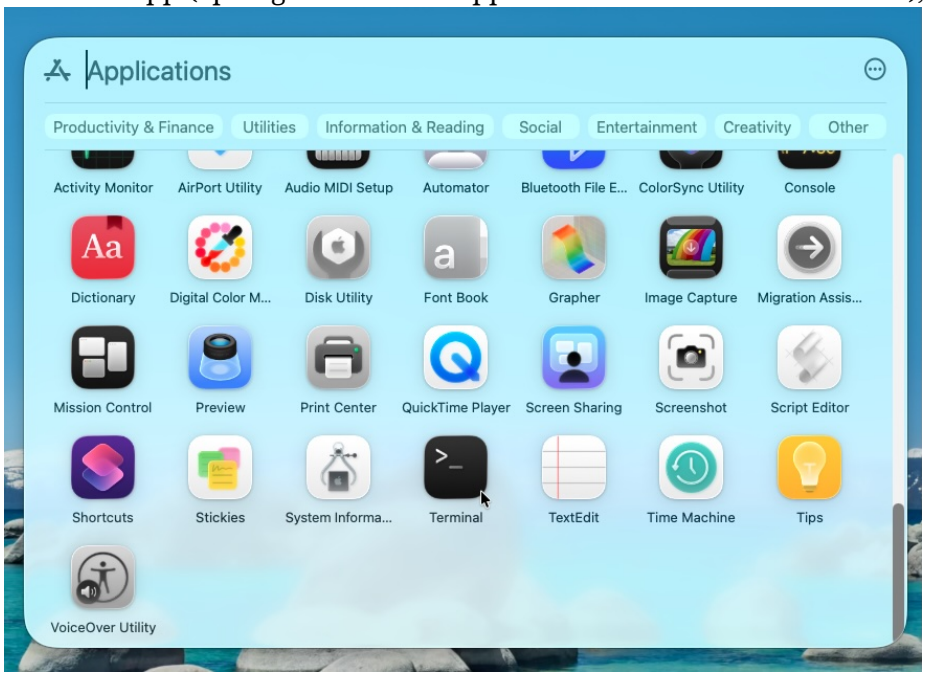
Conda is a tool for managing isolated Python installations ("environments") so that every student in the class runs the *same* package versions, regardless of what's already on your computer. Miniconda (a minimal installer) is preferred to the full Anaconda Distribution to minimize disk space and the chance of conflicting environment dependencies (libraries explicitly or implicitly required by our notebooks invoked using "import" commands), since we'll build our own environment from `environment.yml` anyway. Miniconda is designed to run entirely from the command line (Terminal).

The full Anaconda distribution works fine too if you already have it, and it is recommended on Windows.

🍎 Mac — Option A: Miniconda installation (recommended)

You can install via Terminal or graphically.

Option A1. Install Miniconda from Terminal using brew: Open the Terminal app (Spotlight search or Applications window → “Terminal”),



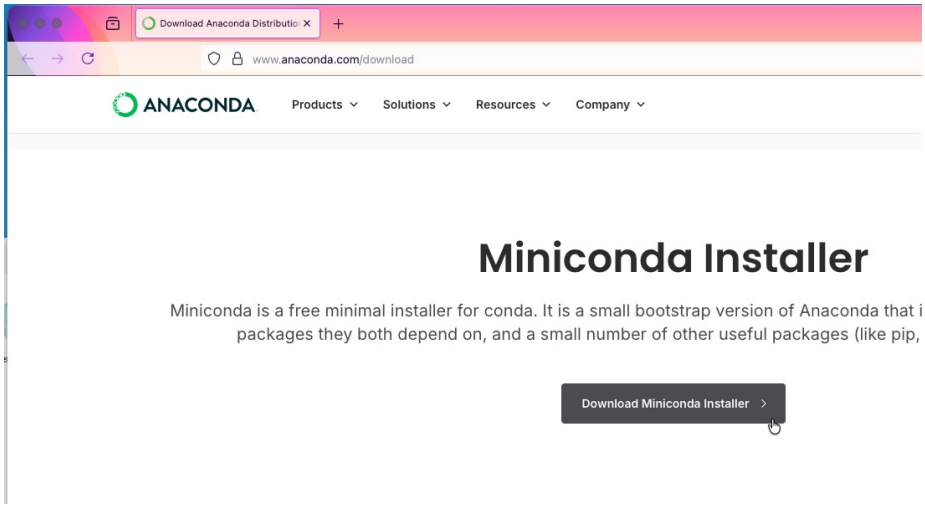
then run:

```
brew install --cask miniconda
```

If you don’t have Homebrew yet, install it first from brew.sh, or install graphically.

Option A2. Graphical installation for Miniconda: Go to anaconda.com/download and download the Miniconda installer. Run the installer and follow the prompts (accept defaults unless you have a reason not to; on the “Install for” screen choose Just Me, not All Users).

Figure 1: Step 1: install conda



🍎 Mac / 🪟 PC — Option B: Graphical Anaconda installer with Navigator application

Go to anaconda.com/download and download the Anaconda installer. Run the installer and follow the prompts (accept defaults unless you have a reason not to; on the “Install for” screen choose Just Me, not All Users).

PC note

On Windows, the installer adds an Anaconda Prompt to your Start Menu — use this instead of the regular Command Prompt / PowerShell for all `conda` or `jupyter` commands below or just use Navigator in this case (unless you checked the box during install to add conda to your system PATH).

Verify the install

Open Terminal (Mac) or Anaconda Prompt (PC) and run:

```
conda --version
```

You should see something like `conda 24.x.x`. If you get "command not found," see Troubleshooting.

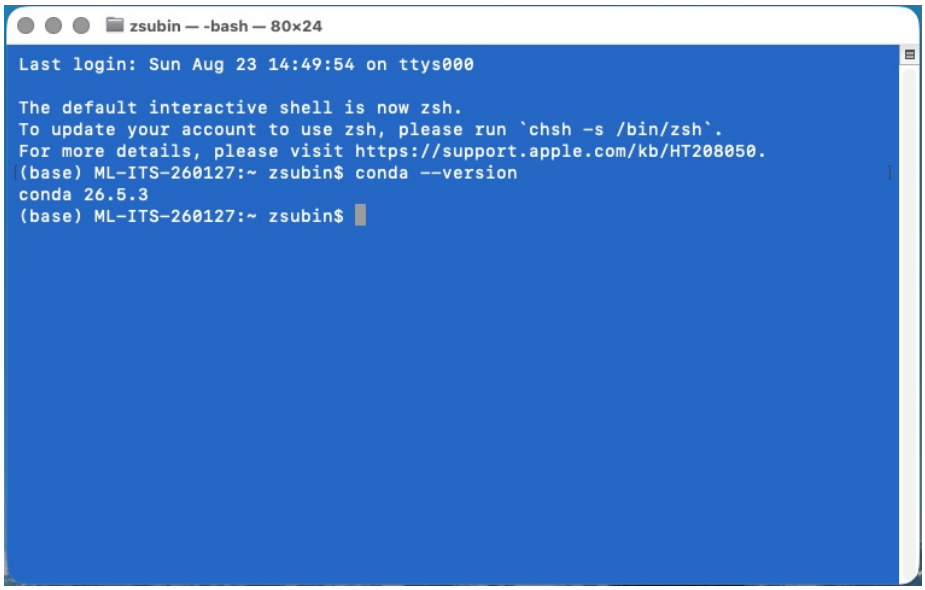


Figure 2: Step 1: Terminal showing successful `conda --version` output

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2. Install GitHub Desktop and sign in

We'll use GitHub Desktop — a graphical app — instead of raw git commands, so you never have to memorize `git` syntax, and to simplify logging into github (you can no longer log in to access private repos from a terminal simply with a username and password; you have to use an ssh-key or OAuth procedure which GitHub Desktop takes care of for you).

1. Download it from desktop.github.com. (You can alternatively install this via brew using the "github" cask.)
2. Install it:
   - 🍎 Mac: open the `.zip`, drag GitHub Desktop into Applications.
   - 🪟 PC: run the installer `.exe`; it installs and launches automatically.
3. Open GitHub Desktop and click Sign in to GitHub.com. This opens your browser to authorize the app — log in with your GitHub account and click Authorize GitHub Desktop, then return to the app.
4. If prompted, confirm your name/email for git commits (your GitHub account name/email is fine).

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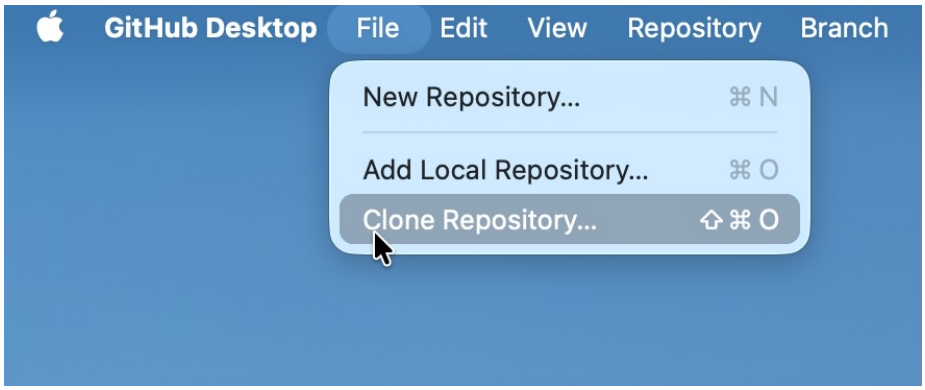
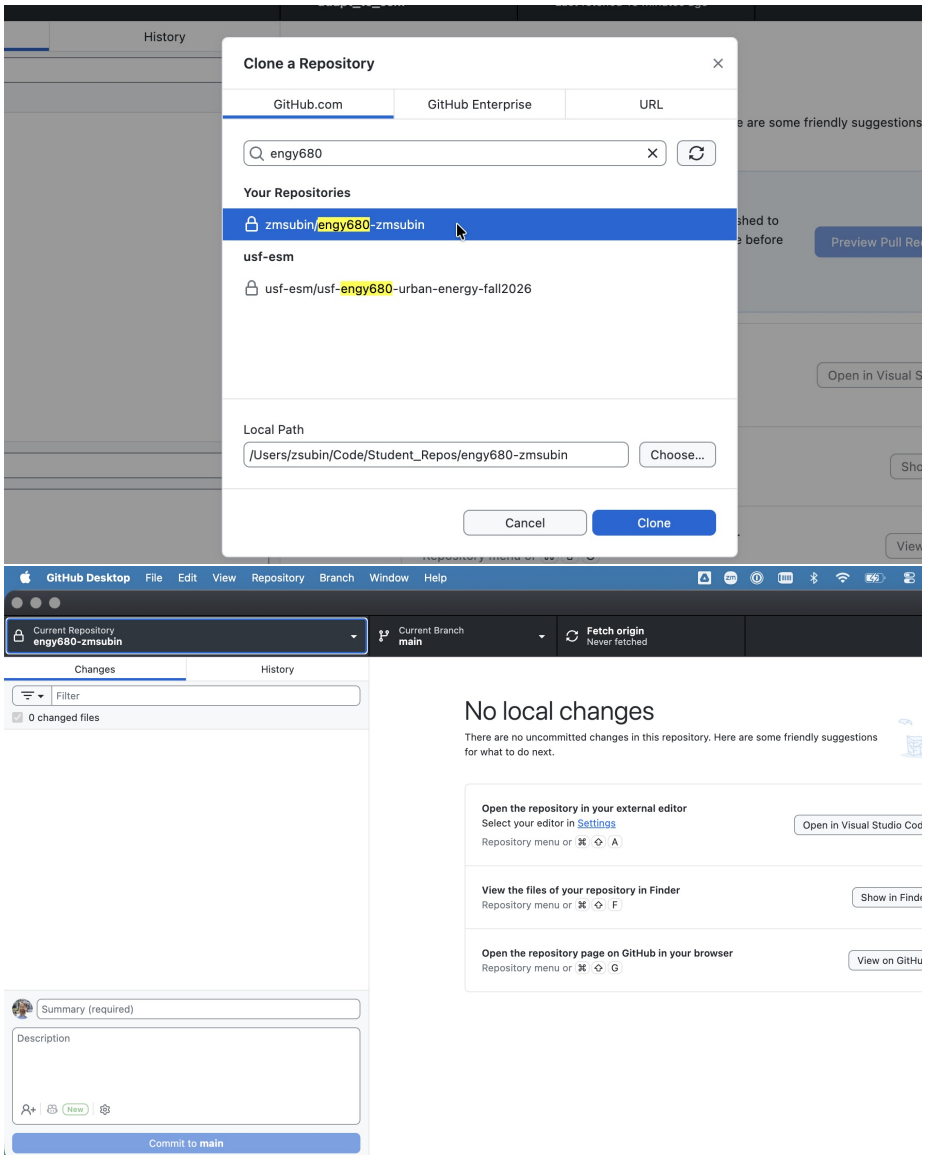
3. Clone your private course repo

Each student has a private repo named `[INSTRUCTOR_USERNAME]/engy680-[YOUR_USERNAME]` (a copy of the shared course repo). 🛑 DO NOT try to do homework in the shared/instructor repo — only your private copy. (The GitHub permissions should be configured to prevent this, but just in case they are not, this will keep your homework private.)

1. In GitHub Desktop, go to File → Clone Repository…
2. Click the GitHub.com tab and find `[INSTRUCTOR_USERNAME]/engy680-[YOUR_USERNAME]` in the list (or paste the URL under the URL tab if it doesn't show up).
3. Choose a Local Path — somewhere easy to find, e.g. `Documents/engy680`.
4. Click Clone and wait for it to finish.





If your repo isn't listed: your invite may not be accepted yet — check your email (including spam) for a GitHub invite, or confirm with the instructor that your GitHub username was received.

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4. Create and activate the conda environment

Your repo includes an `environment.yml` file that lists every Python package the class needs (pandas, jupyterlab, matplotlib, etc.), so everyone has identical versions.

Option A: Command line (Mac or PC)

Open Terminal / Anaconda Prompt, navigate into your cloned repo, and create the environment:

```
cd path/to/engy680
conda env create -f environment.yml -n engy680
```

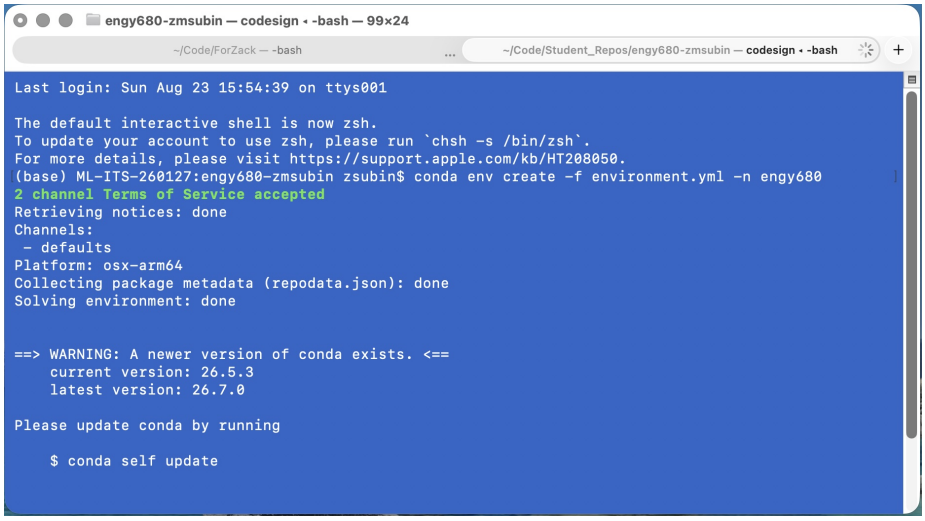


Figure 3: Step 4: installing env

This can take a few minutes ("solving environment…"). When it finishes, activate it:

```
conda activate engy680
```

Your terminal prompt should now show `(engy680)` at the start of the line.

Verify:

```
conda env list
```

You should see `engy680` in the list with an asterisk * next to it if active.

Option B: Anaconda Navigator (graphical, no terminal)

1. Open Anaconda Navigator.

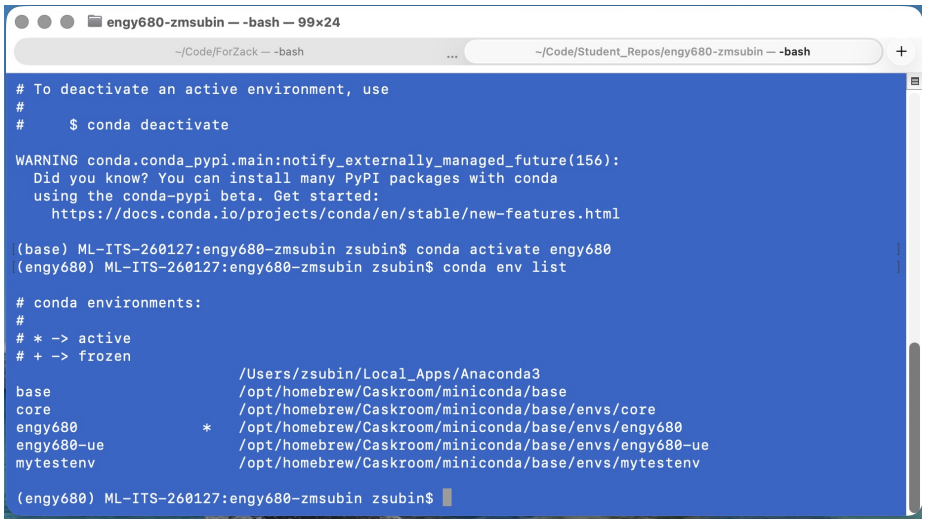
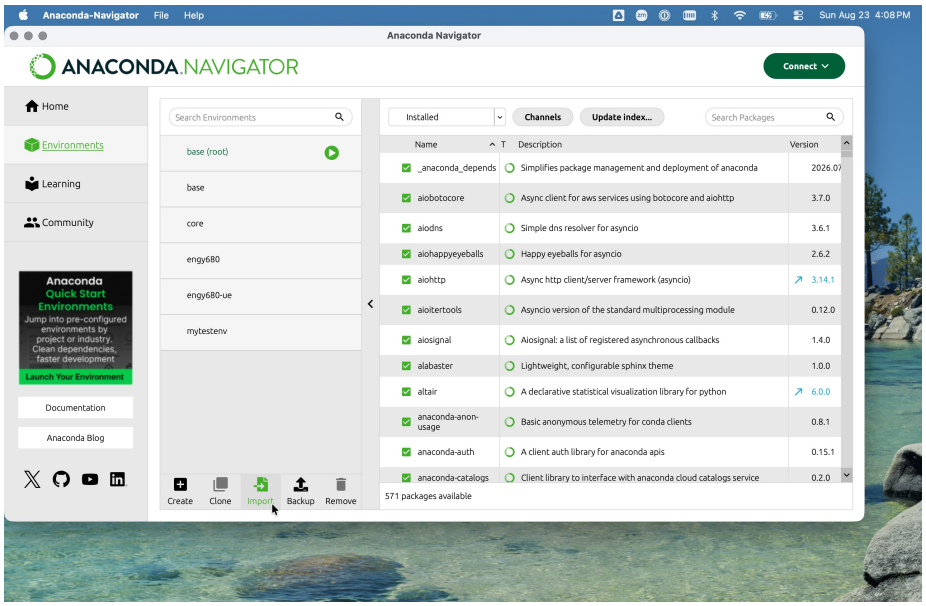
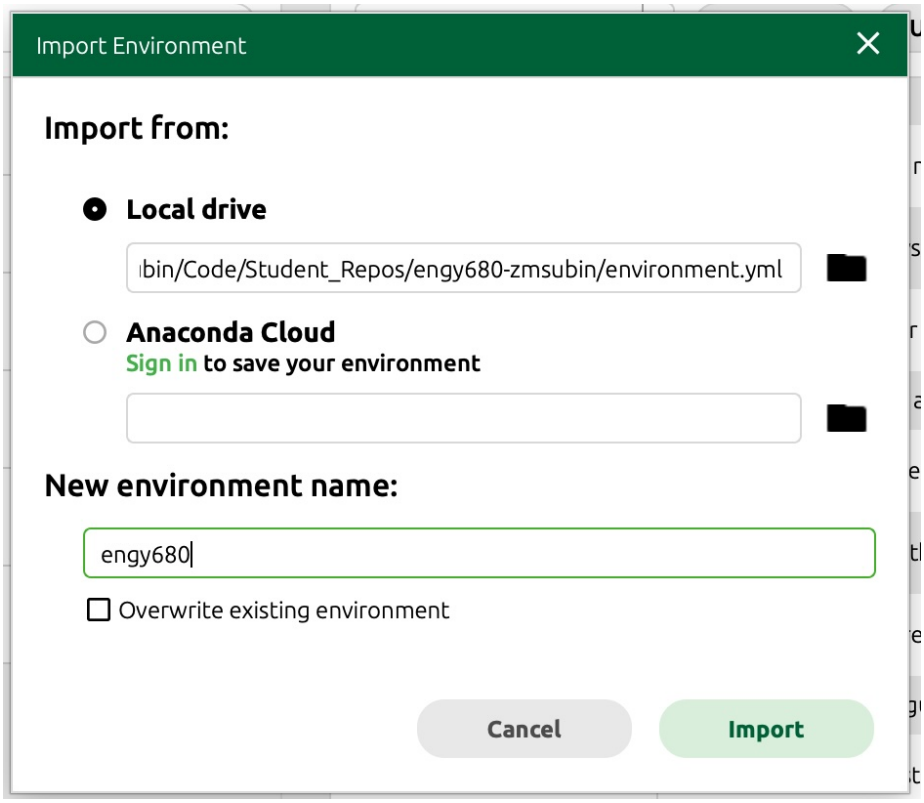


Figure 4: Step 4: successful env

2. Click the Environments tab on the left.
3. Click Import at the bottom of the environments list.
4. Enter a name (e.g. `engy680`), browse to your repo's `environment.yml` under Specification File, and click Import.
5. Wait for the solver — this can take several minutes.





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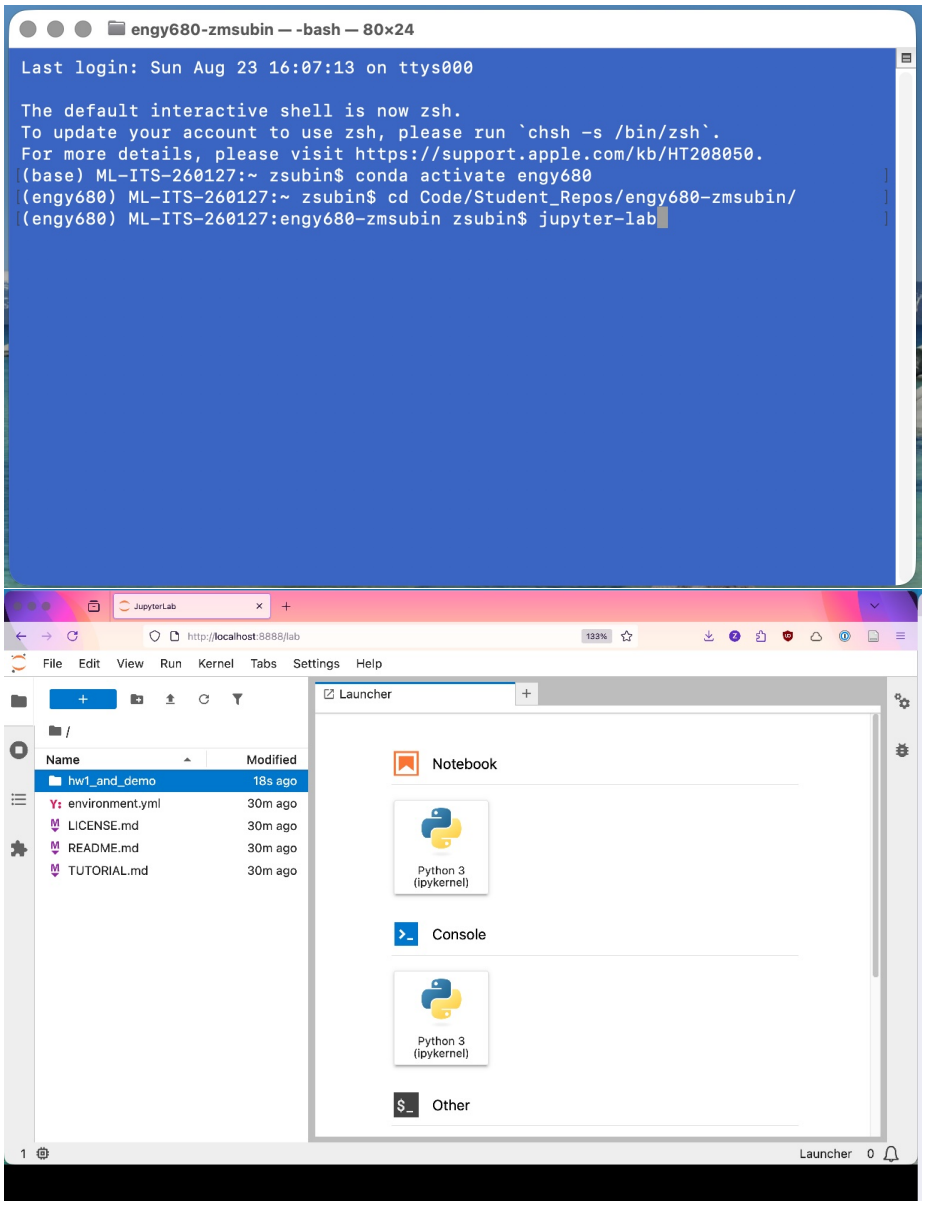
5. Launch Jupyter Lab

With the `engy680` environment active, launch the notebook interface:

Command line

```
conda activate engy680   # if not already active
jupyter-lab
```

A browser tab should open automatically showing the Jupyter Lab file browser. (If not, copy the `http://localhost:8888/...` URL printed in the terminal into your browser.)

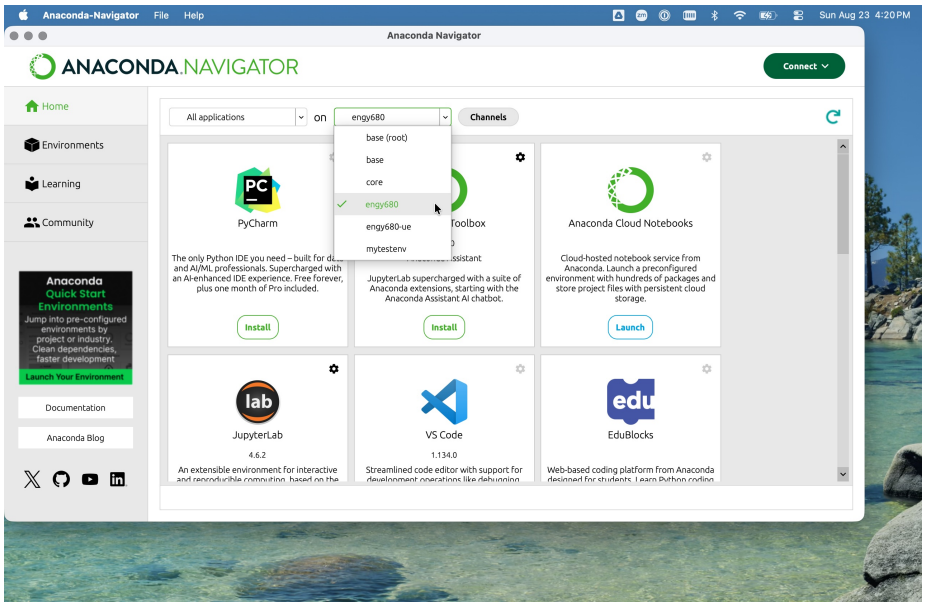


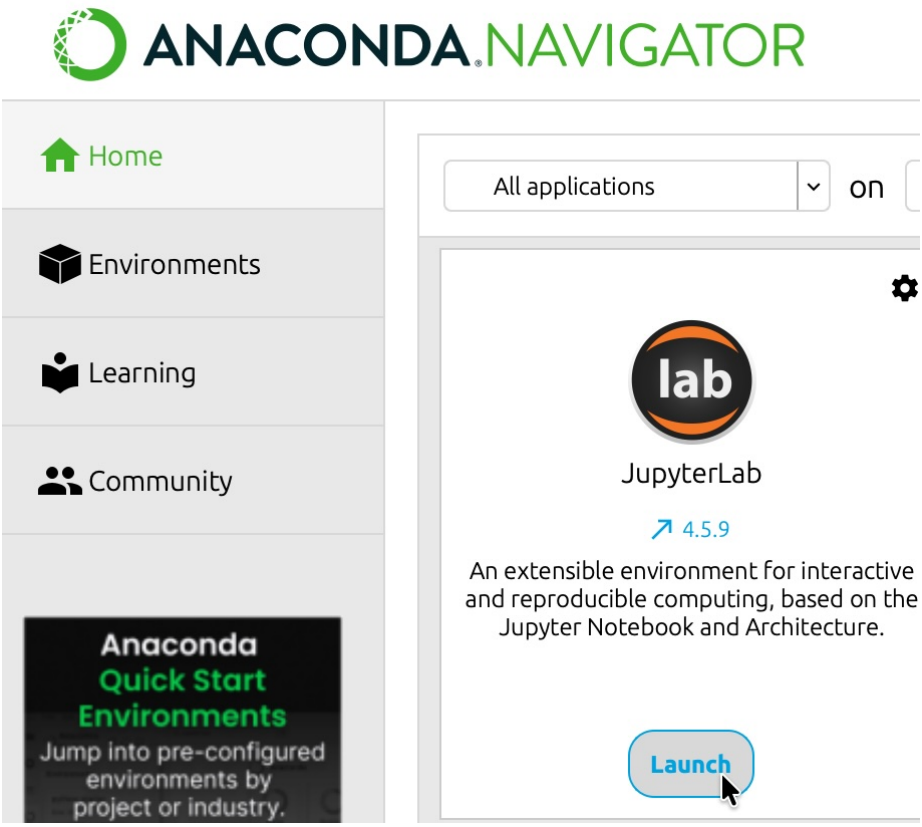
Anaconda Navigator

1. Go to the Home tab.
2. Make sure the environment dropdown at top reads engy680.
3. Click Launch under the JupyterLab tile.

If the JupyterLab tile shows an Install button instead of Launch after you select the `engy680` environment, that usually means `jupyterlab` didn't make it into the environment during Step 4 (e.g., the import was interrupted). Click Install on the tile and wait for it to finish, or re-run the command-line install: `conda activate engy680 && conda install jupyterlab`.

🤔 This step may have problems on some configurations. Since we are using only very standard packages with common usages, you may be able to successfully run the homework using the Base environment that comes with the Navigator package. This may be less somewhat likely to work in subsequent homeworks when we make a map visualization using geopandas.





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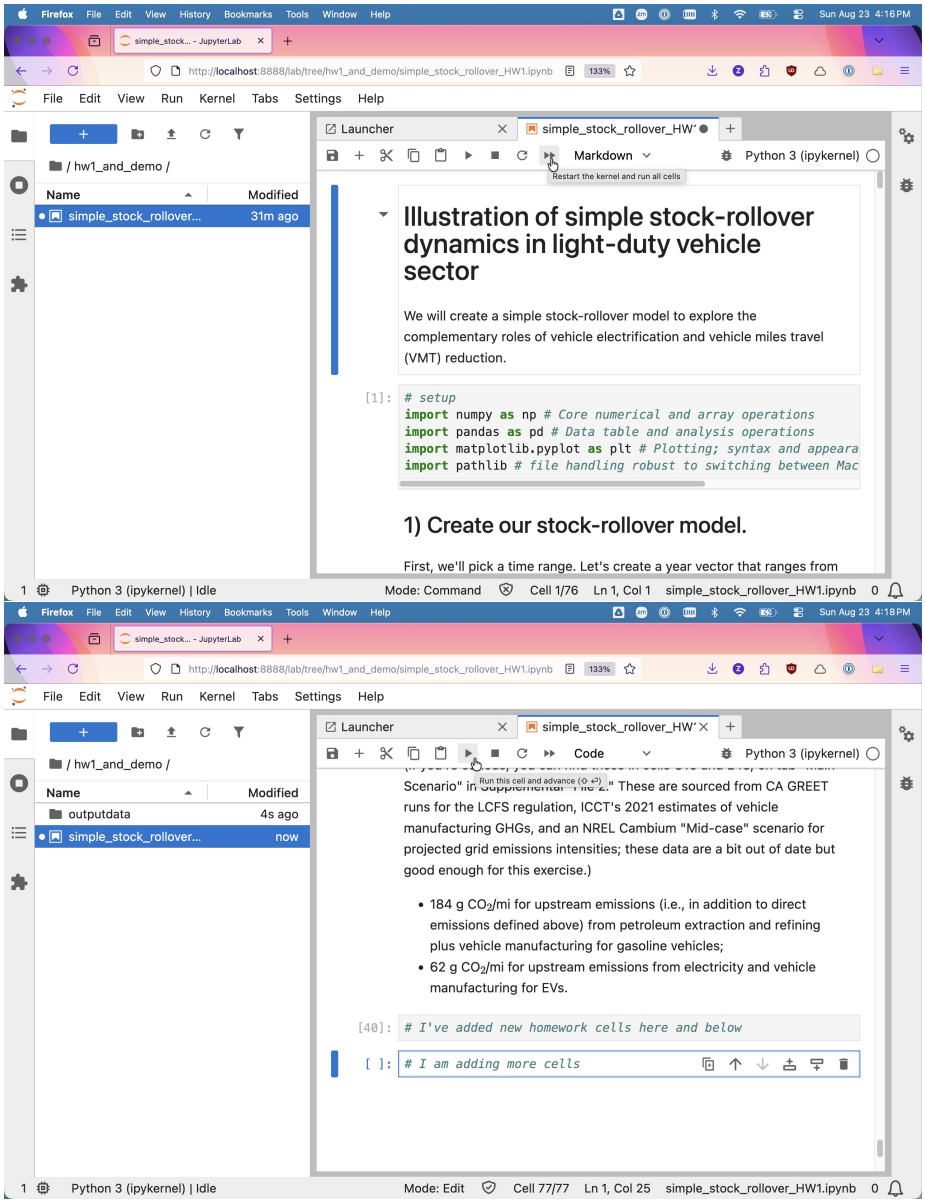
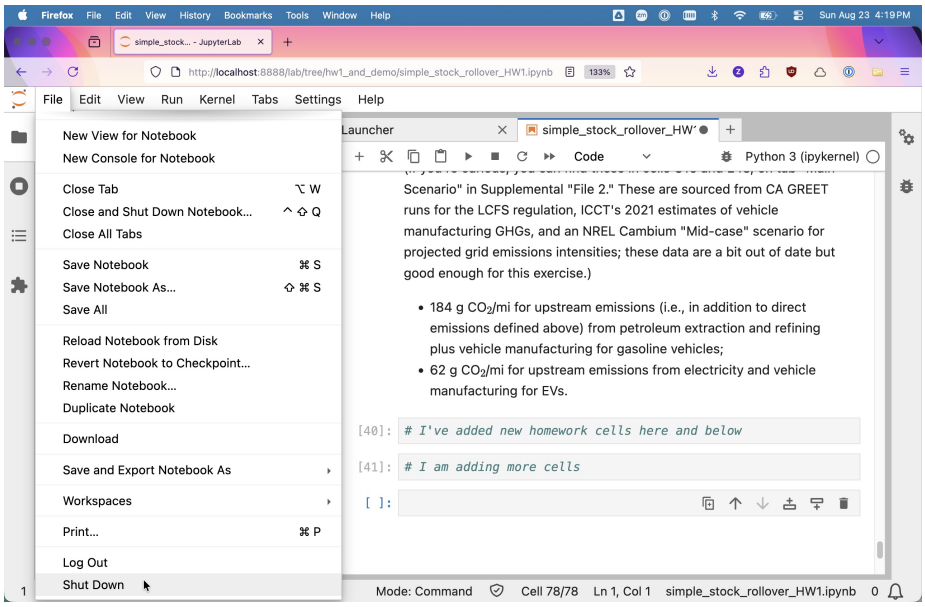
6. Run the demo/homework notebook

1. In the Jupyter Lab file browser (left sidebar), navigate to the homework folder and double-click the notebook file (e.g. `simple_stock_rollover_HW1.ipynb`).
2. Run the existing demo cells top to bottom: Run → Run All Cells (or click each cell and press `Shift+Enter`).
3. Scroll to the bottom, and add new cells for your homework answers:
   - Click the + icon in the toolbar, or press `Esc` then b ("insert cell Below"), to add a new cell.
   - Change a cell to Markdown (dropdown in toolbar) if you want to write text explanations, or leave it as Code.
4. Save frequently: `Cmd/Ctrl + S`, or File → Save Notebook. Jupyter autosaves periodically, but always do a manual save before stepping away.
5. When done, shut down cleanly:
   - File → Shut Down (stops the whole Jupyter Lab server), or
   - Close the browser tab and go back to the terminal, press `Ctrl+C`

twice to stop the server, or click Stop in Navigator.





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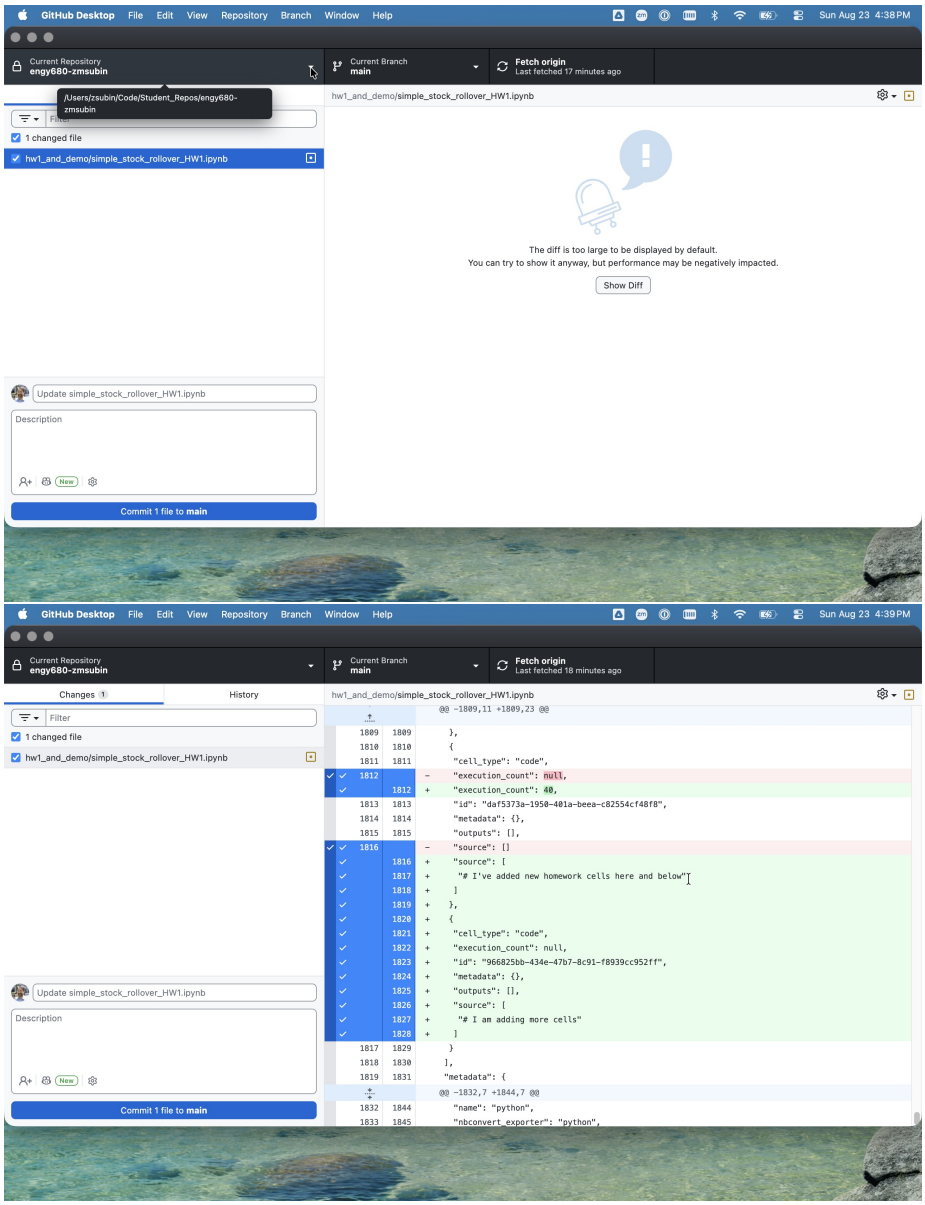
7. Commit and push your changes

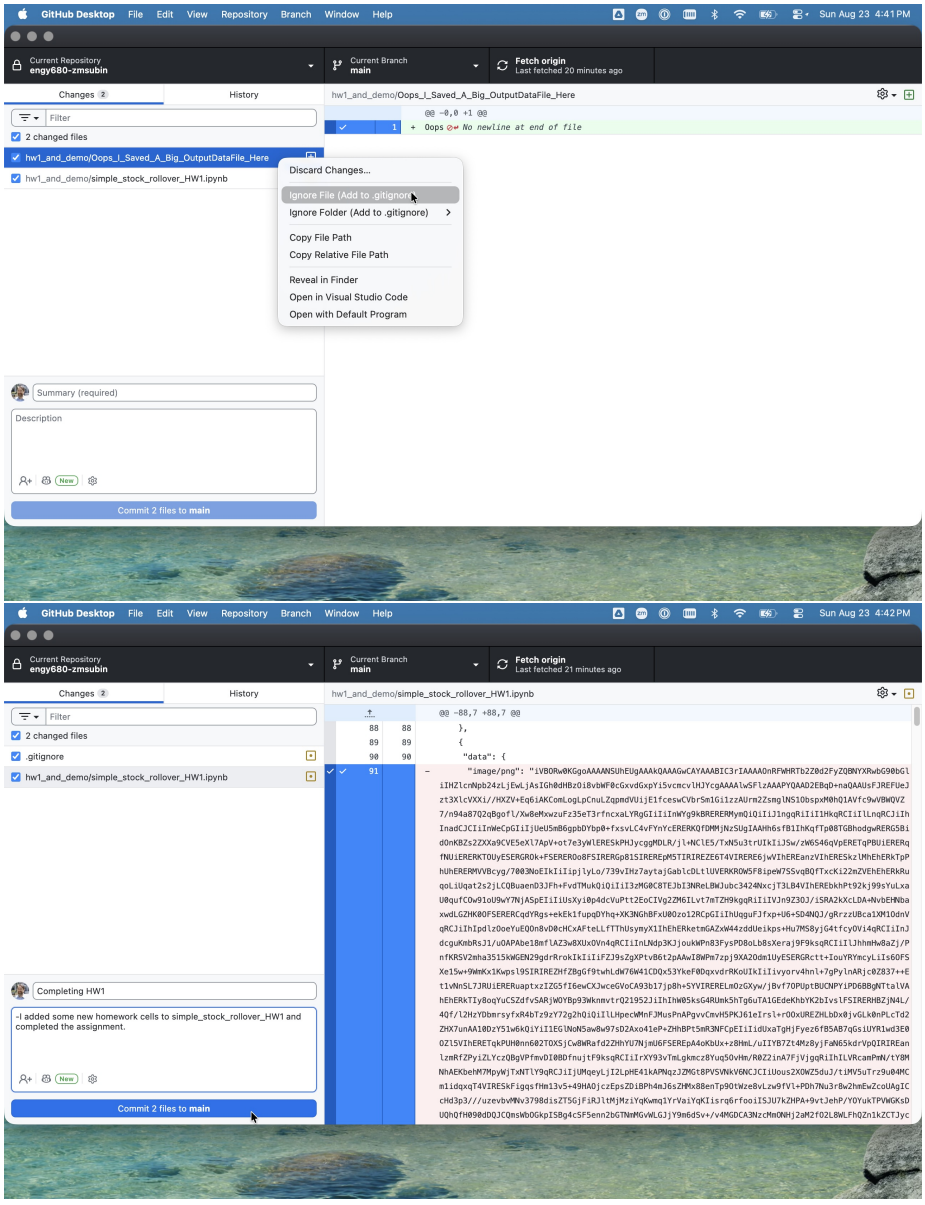
Now save your work back to GitHub using GitHub Desktop.

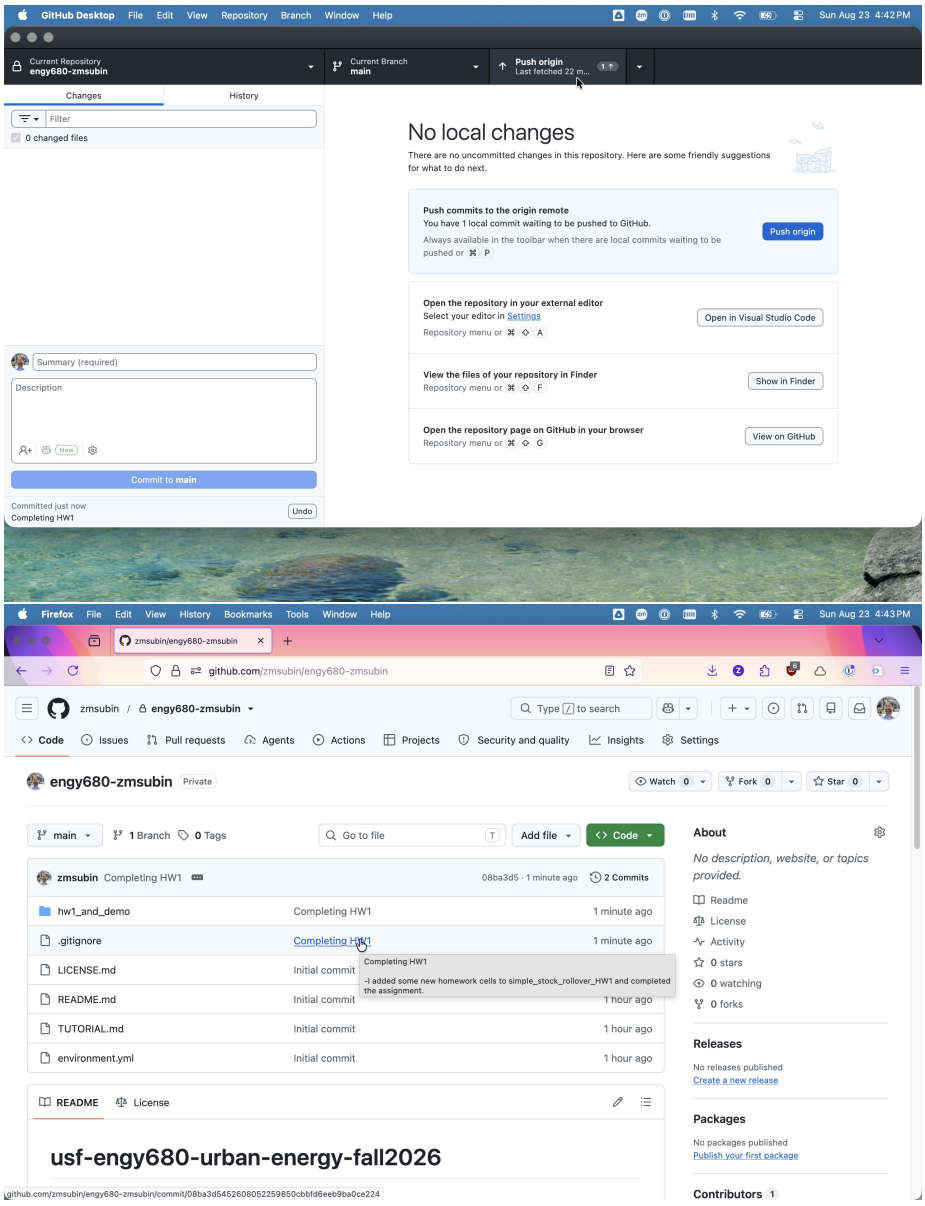
1. Open GitHub Desktop. At the top-left, confirm the current repository dropdown shows `[YOUR_USERNAME]/engy680` — not the shared class repo.
2. Click the Changes tab (left side) to see every file you've modified. Click a file to see its diff (added/removed lines) on the right.

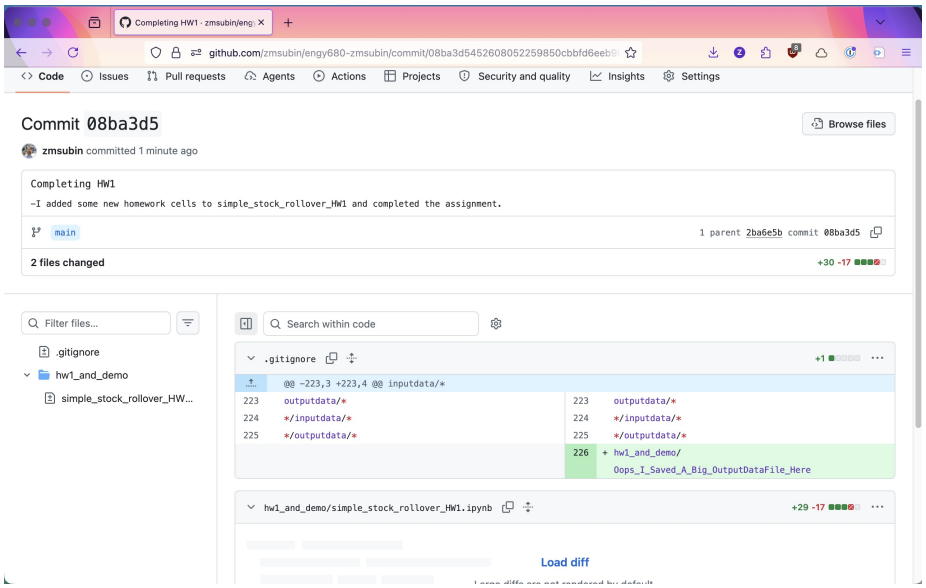
   In contrast to conventional code, (e.g., .py scripts or modules), this diff comparison is not as helpful for Jupyter notebooks which have been saved in a run state because it tracks verbose runtime changes to package versions, binary graphic outputs, etc. Ask about ways to clean this to allow more clean tracking when using notebooks.
3. Check carefully: only your notebook (`.ipynb`) and any code files should be checked/staged, along with any appendix chatbot session markdown files or other small addenda. Uncheck (or right-click → Ignore file, which adds it to `.gitignore`) any large data files, output CSVs, or generated images — these shouldn't go into version control.
4. Write a short Summary (e.g. "Drafting HW1 question 1", "Add HW1 question 1-2 code", "fixed error in question 2 calculation") in the box at bottom left, and optionally a longer Description.

5. Click Commit to main.
6. Click Push origin at the top to upload your commit(s) to GitHub.com.
7. (Optional but reassuring) Open your repo on github.com in a browser and confirm your latest commit appears in the history.




GitHub Desktop File Edit View Repository Branch Window Help
Sun Aug 23 4:41 PM
Current Repository
engy680-zmsubin
Current Branch
main
Fetch origin
Last fetched 20 minutes ago
Changes 2
History
Filter
2 changed files
hw1_and_demo/Oops_I_Saved_A_Big_OutputDataFile_Here
hw1_and_demo/simple_stock_rollover_HW1.ipynb
Discard Changes...
Ignore File (Add to .gitignore)
Ignore Folder (Add to .gitignore)
Copy File Path
Copy Relative File Path
Reveal in Finder
Open in Visual Studio Code
Open with Default Program
@@ −0,0 +1 @@
+ Oops No newline at end of file
Summary (required)
Description
New
Commit 2 files to main
GitHub Desktop File Edit View Repository Branch Window Help
Sun Aug 23 4:42 PM
Current Repository
engy680-zmsubin
Current Branch
main
Fetch origin
Last fetched 21 minutes ago
Changes 2
History
Filter
2 changed files
.gitignore
hw1_and_demo/simple_stock_rollover_HW1.ipynb
hw1_and_demo/simple_stock_rollover_HW1.ipynb
@@ −88,7 +88,7 @@
"image/png": "iVBORw0KGgoAAAANSUhEUgAAAkQAAAGwCAYAAABIC3rIAAAAOnRFWHRTb2Z0d2FyZQBNYXRwbG90bGl
Completing HW1
−I added some new homework cells to simple_stock_rollover_HW1 and completed the assignment.
New
Commit 2 files to main


GitHub Desktop File Edit View Repository Branch Window Help
Sun Aug 23 4:42 PM
Current Repository
engy680-zmsubin
Current Branch
main
Push origin
Last fetched 22 m...
Changes
History
Filter
0 changed files
No local changes
There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.
Push commits to the origin remote
You have 1 local commit waiting to be pushed to GitHub.
Always available in the toolbar when there are local commits waiting to be pushed or ⌘ P
Push origin
Open the repository in your external editor
Select your editor in Settings
Repository menu or ⌘ ⌥ A
Open in Visual Studio Code
View the files of your repository in Finder
Repository menu or ⌘ ⌥ F
Show in Finder
Open the repository page on GitHub in your browser
Repository menu or ⌘ ⌥ G
View on GitHub
Summary (required)
Description
New
Commit to main
Committed just now
Completing HW1
Undo
Firefox File Edit View History Bookmarks Tools Window Help
Sun Aug 23 4:43 PM
zmsubin/engy680-zmsubin
github.com/zmsubin/engy680-zmsubin
zmsubin / engy680-zmsubin
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Security and quality
Insights
Settings
engy680-zmsubin
Private
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
T
Add file
Code
zmsubin Completing HW1
08ba3d5 · 1 minute ago
2 Commits
hw1_and_demo
Completing HW1
1 minute ago
.gitignore
Completing HW1
1 minute ago
LICENSE.md
Initial commit
README.md
Initial commit
1 hour ago
TUTORIAL.md
Initial commit
1 hour ago
environment.yml
Initial commit
1 hour ago
Completing HW1
-I added some new homework cells to simple_stock_rollover_HW1 and completed the assignment.
README
License
usf-engy680-urban-energy-fall2026
github.com/zmsubin/engy680-zmsubin/commit/08ba3d5452608052259850cbbfd6eeb9ba0ce224
About
No description, website, or topics provided.
Readme
License
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1



💡 Reminder: You don't need to save copies like `HW1_v2.ipynb` or `HW1_final.ipynb` — that's what git/version history is for. Just keep committing to the same file as you make progress.

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8. Troubleshooting

Symptom	Likely cause / fix
`conda: command not found`	Restart your terminal after installing, or (Mac) run `conda init zsh` (or `bash`) then restart terminal. On PC, use the Anaconda Prompt, not regular Command Prompt.
`conda env create` is very slow / hangs on "Solving environment"	Normal — can take 5–15 minutes on first run. If it truly hangs (30+ min), try `conda install -n base conda-libmamba-solver` then re-run with `--solver=libmamba`.
GitHub Desktop can't find your repo when cloning	Confirm you accepted the email invite / your GitHub username was sent to the instructor. Try refreshing (sign out/in) in GitHub Desktop.

Symptom	Likely cause / fix
Jupyter opens but can't find installed packages (`ModuleNotFoundError`)	You likely launched Jupyter from the wrong environment. Confirm the terminal prompt shows `(engy680)`, or in Navigator confirm the environment dropdown is set correctly before clicking Launch.
GitHub Desktop won't let you push (file too large)	GitHub has a ~100 MB per-file hard limit. Un-stage the large file, delete it from tracking, add it to `.gitignore`, and re-commit.
Mac: "cannot be opened because the developer cannot be verified"	Right-click the app → Open (instead of double-click) the first time, then confirm.